Spectral Exponents of the Twelve-Lead ECG Reveal the Anatomy of Cardiac Conduction Disorders and a Bifurcation Between Aging and Disease

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Abstract

The aperiodic ($1/f^{\beta}$) component of the electrocardiogram carries information about broadband cardiac electrical dynamics, yet it remains largely unexplored as a quantitative marker of conduction system integrity. Here we apply Irregular-Resampling Auto-Spectral Analysis (IRASA) to 412,730 twelve-lead ECG recordings across three datasets spanning three continents—PTB-XL (Germany, $n = 21{,}799$), Chapman-Shaoxing (China, $n = 45{,}152$), and CODE-15% (Brazil, $n = 345{,}779$)—to extract the spectral exponent β, a single number per lead quantifying the balance between order and disorder in the cardiac electrical signal. We report five findings. First, each diagnostic category deviates from the healthy reference in a specific direction and magnitude, with conduction disturbances showing the strongest divergence (Cohen's $d = 0.99$). Second, the twelve-lead β-vector encodes the spatial anatomy of the conduction system: a classifier distinguishes complete left from complete right bundle branch block with AUC = 0.982 on the discovery cohort (PTB-XL), and this result replicates exactly on two independent external cohorts (AUC = 0.982 on Chapman-Shaoxing; AUC = 0.979 on CODE-15%), despite substantial variation in absolute β-values across recording equipment ($\beta_{\mathrm{NORM}} = 1.52$–$2.75$). Third, within a curated healthy cohort (PTB-XL), aging and pathology drive β in opposite directions—healthy aging flattens the spectrum (β decreases, $\rho = -0.179$) while overt conduction disease steepens it (β increases), creating a bifurcation from the healthy operating point. Beat-to-beat morphological variability independently confirms age-related complexity loss ($\rho = 0.165$, $p < 10^{-19}$). Fourth, among ECGs labeled normal by cardiologists, β distinguishes truly clean recordings from those with subclinical conduction findings (AUC = 0.754), with the direction of β-shift inverted relative to overt disease. Fifth, β does not independently predict all-cause mortality beyond established diagnoses (adjusted HR = 1.02, $p = 0.83$ in CODE-15%), establishing it as a diagnostic marker of conduction anatomy rather than an independent prognostic biomarker. These results demonstrate that the spectral exponent captures a cross-population invariant of cardiac electrophysiology: the geometry of the twelve-lead β-vector is determined by conduction system anatomy and is reproducible across equipment, ethnicity, and geography. **Keywords:** electrocardiography, spectral exponent, $1/f$ noise, IRASA, cardiac conduction, aging, PTB-XL, external validation, cross-population invariance

1 Introduction

The electrocardiogram (ECG) has been the primary tool for cardiac diagnosis for over a century. Clinical interpretation focuses on periodic features—heart rate, P-wave morphology, QRS duration, ST-segment deviation, QT interval—that reflect specific phases of the cardiac cycle. Deep learning approaches have extended this further, achieving diagnostic performance exceeding that of expert cardiologists by extracting complex morphological patterns from the full waveform [Strodthoff et al., 2021, Ribeiro et al., 2020]. Yet the ECG signal also contains an aperiodic component: a broadband $1/f^{\beta}$ background upon which periodic cardiac oscillations are superimposed. This aperiodic component, characterized by the spectral exponent β, reflects the statistical structure of electrical fluctuations across all frequencies simultaneously. In neurophysiology, the analogous parameter in EEG/MEG signals has been extensively studied and linked to excitation-inhibition balance [Gao et al., 2017], arousal [Lendner et al., 2020], and aging [Voytek et al., 2015]. In cardiology, however, the focus has been almost exclusively on heart rate variability (HRV)—the $1/f$ scaling of beat-to-beat interval fluctuations—rather than on the spectral properties of the ECG waveform itself. The HRV literature has established that healthy cardiac dynamics exhibit complex, multiscale fluctuations near a critical state, while aging and disease degrade this complexity [Goldberger et al., 2002]. Detrended fluctuation analysis of RR-interval series reveals that healthy hearts show scaling exponents near $\alpha \approx 1.0$ (equivalent to $1/f$ noise), with systematic deviations in heart failure, atrial fibrillation, and aging [Peng et al., 1995, Ivanov et al., 1999]. Lipsitz & Goldberger [1992] formalized this as the "loss of complexity" hypothesis: healthy physiology is characterized by complex, multiscale dynamics, while disease and aging produce a degradation toward either excessive regularity or uncorrelated randomness. Whether the same principles apply to the ECG waveform—the voltage signal itself, not the derived RR-interval series—remains largely untested. The ECG waveform reflects the collective depolarization and repolarization of approximately two billion cardiomyocytes, and its spectral properties are determined by the quality of electrical coupling (gap junctions), conduction velocity (ion channels), and tissue architecture (fibrosis, scarring). A handful of studies have applied multifractal analysis to ECG voltage signals [Yang et al., 2020, Shekatkar et al., 2017] or examined ECG spectral slopes in the context of aging [Schmidt et al., 2025], but no systematic analysis has mapped the spectral exponent across diagnostic categories in a large clinical dataset. A key methodological challenge is that the ECG contains powerful periodic components—QRS harmonics extending to the 15th harmonic of heart rate—that contaminate spectral slope estimation if not properly separated. Schmidt et al. [2025] identified a spectral knee near 15 Hz and employed IRASA (Irregular-Resampling Auto-Spectral Analysis; Wen & Liu 2016) to separate periodic and aperiodic components. We adopt the same approach on a substantially larger dataset with clinical diagnostic labels. Here we analyze the PTB-XL dataset [Wagner et al., 2020]—21,799 ten-second, twelve-lead ECG recordings from 18,869 patients with expert diagnostic labels spanning five superclasses and 23 subclasses—and externally validate on two independent datasets totaling over 390,000 additional recordings. We extract the spectral exponent β from the aperiodic component of each lead using IRASA and investigate five questions: (1) Does β differ systematically across diagnostic categories? (2) Can the twelve-lead β-vector discriminate conduction disorder subtypes and reconstruct lesion anatomy? (3) How does β change with healthy aging, and how does this relate to disease-related changes? (4) Can β detect subclinical conduction abnormalities in recordings labeled normal? (5) Does β independently predict mortality?

2 Methods

2.1 Discovery Dataset

We used the PTB-XL electrocardiography dataset [Wagner et al., 2020], version 1.0.3, accessed via PhysioNet and Kaggle. The dataset contains 21,799 ten-second, twelve-lead ECG recordings sampled at 500 Hz from 18,869 unique patients (52% male, median age 62 years, range 2–89; one outlier with age = 300 excluded). Each recording was independently annotated by up to two cardiologists with diagnostic labels organized into five superclasses: Normal (NORM, $n = 9{,}514$), Myocardial Infarction (MI, $n = 5{,}469$), ST/T Changes (STTC, $n = 5{,}235$), Conduction Disturbance (CD, $n = 4{,}898$), and Hypertrophy (HYP, $n = 2{,}649$). Recordings may carry multiple labels. Within CD, subclasses include Complete Left Bundle Branch Block (CLBBB, $n = 536$), Complete Right Bundle Branch Block (CRBBB, $n = 537$), Incomplete Right Bundle Branch Block (IRBBB, $n = 1{,}102$), Left Anterior Fascicular Block (LAFB, $n = 1{,}604$), First-Degree AV Block (1AVB, $n = 779$), and Intraventricular Conduction Delay (IVCD, $n = 780$). Subclass counts reflect recordings with $\geq 80\%$ annotator confidence. We used the predefined stratified folds for train-test splits, with folds 9–10 reserved for testing.

2.2 Signal Preprocessing

Each twelve-lead ECG was preprocessed as follows: (1) bandpass filtering at 0.5–100 Hz (4th-order Butterworth) to remove baseline drift and high-frequency noise; (2) notch filtering at 50 Hz (2nd-order IIR) to remove powerline interference.

2.3 Spectral Exponent Extraction via IRASA

We extracted the aperiodic spectral exponent β using IRASA [Wen & Liu, 2016]. IRASA exploits the fact that resampling a signal at an irrational rate h shifts periodic (oscillatory) components in the frequency domain but preserves the power-law structure of the aperiodic component. By computing the power spectral density (PSD) at multiple resampling factors ($h = 1.1, 1.15, \ldots, 1.9$) and taking the geometric mean of each $(h, 1/h)$ pair, oscillatory peaks are attenuated while the aperiodic background is preserved. For each lead of each recording, we: (1) computed the PSD using Welch's method (2-second Hanning windows, 50% overlap); (2) applied IRASA with 17 resampling factors to obtain the fractal (aperiodic) PSD; (3) performed linear regression of $\log_{10}(\text{Power})$ on $\log_{10}(\text{Frequency})$ in the range 2–45 Hz to obtain the spectral exponent β and the coefficient of determination R^2. This procedure yields, for each recording, a 12-dimensional β-vector $[\beta_{\text{I}}, \beta_{\text{II}}, \ldots, \beta_{\text{V6}}]$ and a corresponding R^2-vector.

2.4 Derived Metrics

From the 12-dimensional β-vector, we computed: β_{mean} (mean across all 12 leads), σ_β (standard deviation across leads), regional means (β_{anterior} for V1–V4, β_{lateral} for I, aVL, V5, V6, β_{inferior} for II, III, aVF), and beat-to-beat morphological variability (defined as $1 -$ mean correlation of individual beats wit

2.5 Statistical Analysis

Group comparisons used the Kruskal-Wallis test with post-hoc Mann-Whitney U tests and Bonferroni correction. Effect sizes are reported as Cohen's d. Age-related trends were assessed by Spearman correlation and segmented regression with breakpoint estimation (Davies test). ANCOVA with age, sex, and recording device as covariates confirmed diagnostic effects.

2.6 Classification

We used logistic regression, random forest (500 trees), and gradient boosting machine (GBM, 200 trees) with features consisting of per-lead β and R^2 values (24 features), regional means and σ_β (6 features), and demographics (age, sex). Performance was evaluated by AUC-ROC on held-out test folds.

2.7 External Validation Datasets

Chapman-Shaoxing [Zheng et al., 2020]: 45,152 twelve-lead ECGs at 500 Hz from Shaoxing People's Hospital, China (GE MUSE system). Labels include CLBBB ($n = 213$) and CRBBB ($n = 1{,}096$). **CODE-15%** [Ribeiro et al., 2020]: 345,779 twelve-lead ECGs at 400 Hz from the Telehealth Network of Minas Gerais, Brazil. Labels include LBBB ($n = 414$) and RBBB ($n = 557$). Mortality follow-up data available. IRASA frequency range adjusted to 2–40 Hz. For both external datasets, preprocessing and β-extraction were identical to PTB-XL except that CODE-15% used a 60 Hz notch filter (Brazilian powerline standard). The lower sampling rate of CODE-15% (400 Hz vs. 500 Hz) reduces the Nyquist frequency from 250 to 200 Hz, but our fitting range (2–40 Hz for CODE-15%, 2–45 Hz for 500 Hz datasets) lies well within the resolved bandwidth. The spectral exponent β is extracted from the aperiodic component in this low-frequency range and is not affected by the difference in Nyquist frequency.

2.8 Subclinical Conduction Findings

To test whether β detects subclinical conduction abnormalities, we identified recordings labeled NORM ($\geq 80\%$ confidence) that also carried low-confidence annotations for conduction-related codes. Specifically, we defined "subclinical" as any NORM recording co-annotated with one or more of: IRBBB ($n = 209$), IVCD ($n = 80$), 1AVB ($n = 23$), LAFB ($n = 14$), or LPFB ($n = 6$), totaling 328 recordings. The "pure NORM" control group consisted of 179 recordings with NORM as the sole annotation.

2.9 Mortality Analysis

In CODE-15%, we assessed whether β independently predicts all-cause mortality using Cox proportional hazards regression at three levels of adjustment: (1) unadjusted; (2) age and sex; (3) age, sex, and all diagnostic labels.

3 Results

3.1 The β-Landscape of Cardiac Diagnoses

The healthy heart operates at a characteristic spectral exponent $\beta_{\text{mean}} = 1.76 \pm 0.34$. Each diagnostic superclass shows a distinct β-profile (Kruskal-Wallis $H = 649$, $p < 10^{-139}$; Figure 1a). Conduction Disturbance (CD) is the strongest outlier, with $\beta_{\text{mean}} = 2.15 \pm 0.63$ (Cohen's $d = 0.99$ vs. NORM, $p < 10^{-124}$). Myocardial Infarction (MI, $d = 0.28$) and Hypertrophy (HYP, $d = 0.29$) show moderate elevations. ST/T Changes do not differ significantly from NORM ($d = 0.02$), consistent with STTC reflecting repolarization rather than conduction abnormalities. ANCOVA confirmed that the diagnostic effect is not confounded by demographics: $F = 333$, $p < 10^{-276}$, partial $\eta^2 = 0.076$. The twelve-lead β-vector reveals spatial specificity (Figure 1c). CLBBB produces extreme β-elevation in V1–V4 ($\Delta\beta$ up to $+1.47$ in V3), corresponding to the anatomical course of the left bundle branch through the interventricular septum. CRBBB shows moderate elevation primarily in V1 ($\Delta\beta = +0.49$).

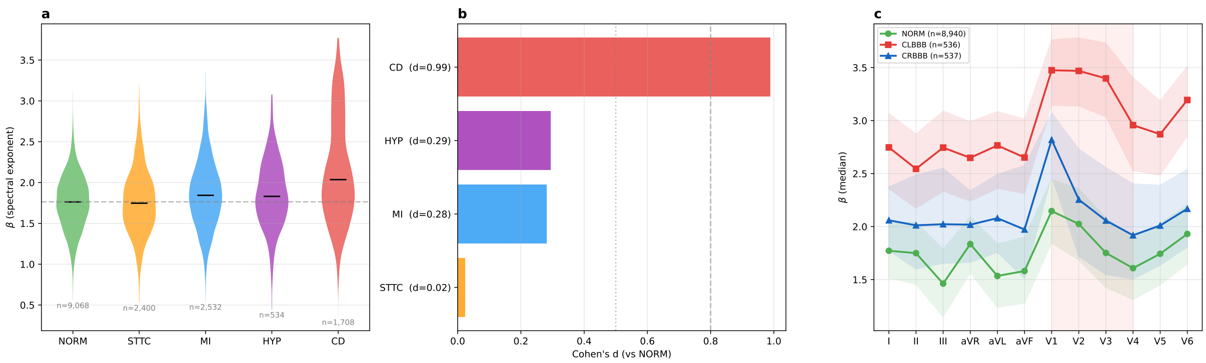


Figure 1: **The β-landscape of cardiac diagnoses.** (a) Violin plots of β_{mean} for each diagnostic superclass. Horizontal dashed line marks $\beta = 1.76$. CD is the strongest outlier (Cohen's $d = 0.99$). (b) Cohen's d effect sizes vs. NORM. (c) Spatial fingerprint: median β per lead for NORM, CLBBB, and CRBBB. Shaded region marks precordial leads V1–V4.

3.2 Spectral Anatomy of Conduction Disorder Subtypes

A multiclass classifier distinguishing six CD subtypes achieved macro-averaged AUC $= 0.862$ (Figure 2b). Per-subtype AUCs ranged from 0.974 (CLBBB) to 0.774 (IVCD). The pairwise CLBBB vs. CRBBB classification achieved AUC $= 0.982$ (Figure 2c). For context, deep learning on raw waveforms achieves AUC ≈ 0.99 using $\sim$60,000 input features [Strodthoff et al., 2021]. Our approach reaches 98.2% of that performance with 54 interpretable features. Radar plots of $\Delta\beta$ per lead for each subtype (Figure 2e) directly map lesion anatomy: CLBBB shows dominant elevation in V1–V4 (left bundle through the septum); CRBBB shows focal elevation in V1 (right bundle proximity); LAFB shows elevation centered on aVL (anterior fascicle to lateral wall); 1AVB shows minimal spatial signature (AV-node pathology produces diffuse delay). Feature importance analysis confirms anatomical specificity: R^2 in V1 is the single most important feature (importance = 0.298).

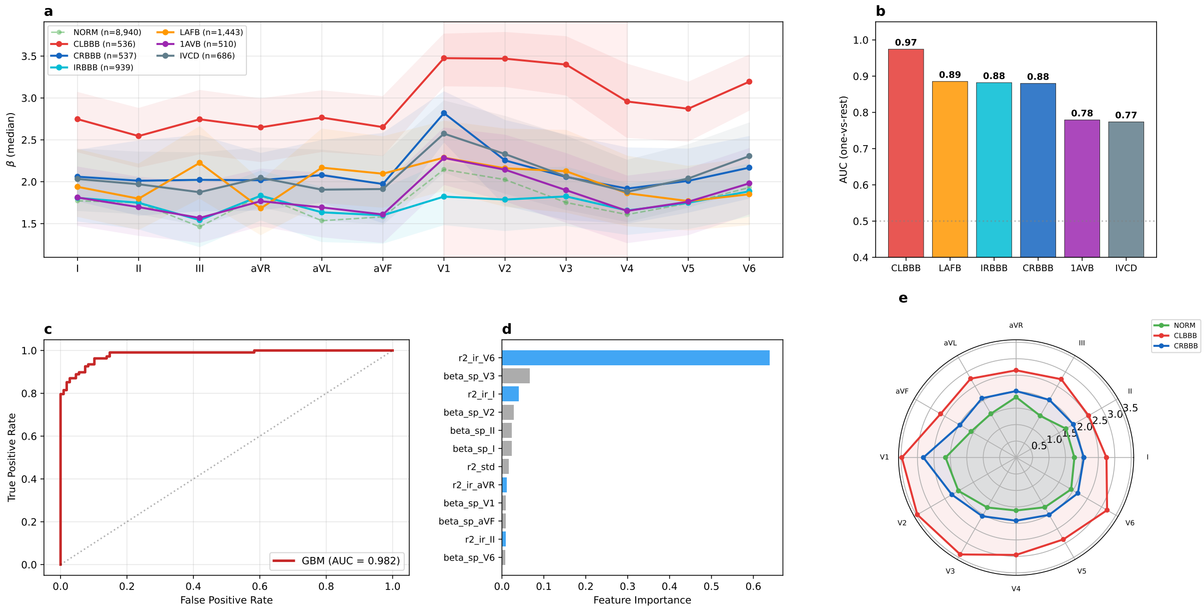


Figure 2: **Spectral anatomy of conduction disorder subtypes.** (a) Spatial fingerprints of six CD subtypes. (b) Per-subtype AUC (one-vs-rest). (c) ROC for CLBBB vs. CRBBB (AUC = 0.982). (d) Feature importance for CLBBB vs. CRBBB classifier. (e) Radar plot of β per lead.

3.3 Aging Bifurcation: Opposite Trajectories of Age and Disease

Among 8,940 NORM recordings, β_{mean} declines with age (Spearman $\rho = -0.179$, $p < 10^{-65}$; Figure 3a). Women show higher baseline β but steeper decline ($\rho_F = -0.22$ vs. $\rho_M = -0.14$). This direction is opposite to overt pathology: CD and MI shift β *upward*, while aging shifts β *downward* (Figure 3c). The healthy operating point sits at a bifurcation between two modes of complexity loss: (1) aging-related spectral flattening (micro-fragmentation of conduction) and (2) disease-related spectral steepening (large-scale conduction block). Segmented regression identifies a breakpoint at age 53 ($F = 17.8$, $p = 1.9 \times 10^{-8}$; Figure 3b), after which spatial heterogeneity (σ_β) accelerates. Beat-to-beat morphological variability independently increases with age ($\rho = 0.165$, $p < 10^{-19}$).

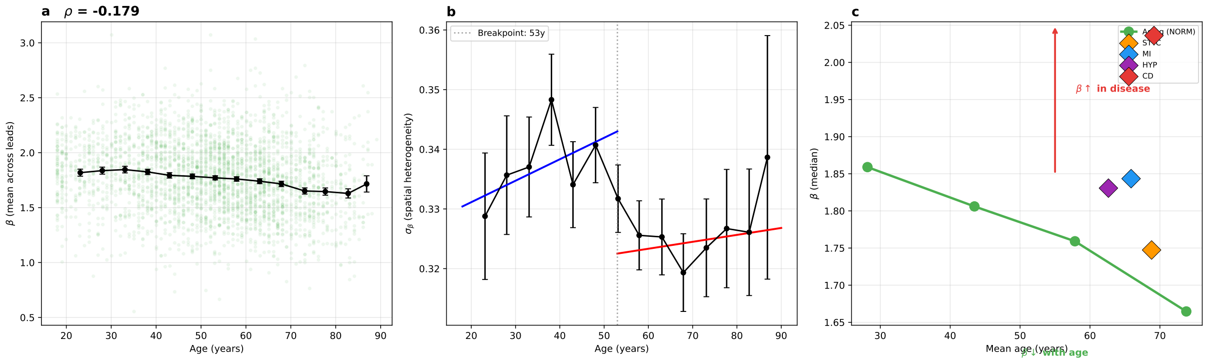


Figure 3: **The aging bifurcation.** (a) β_{mean} vs. age in 8,940 healthy recordings ($\rho = -0.179$). (b) Segmented regression of σ_β with breakpoint at age 53. (c) Bifurcation diagram: aging ($\beta \downarrow$) vs. pathology ($\beta \uparrow$) from the healthy operating point.

3.4 Subclinical Detection

Among NORM recordings, 179 had no conduction findings and 328 had subclinical findings. A classifier achieved AUC $= 0.754$ (Figure 4c). The strongest feature was β_{anterior} (Cohen's $d = -0.551$; Figure 4b), with subclinical recordings showing *lower* β—inverted relative to overt CD. Three regimes form a continuum: subclinical fragmentation ($\beta \downarrow$), normal ($\beta \approx 1.76$), overt blockade ($\beta \uparrow$).

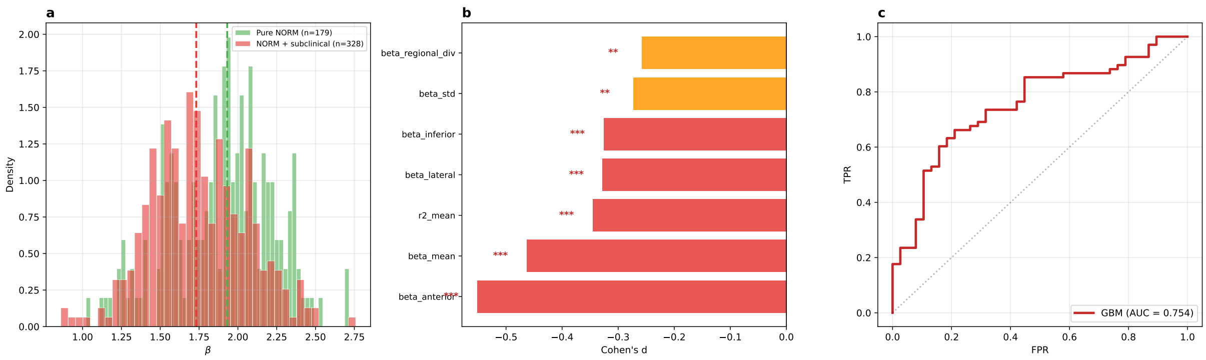


Figure 4: **Subclinical detection.** (a) β_{mean} distributions: pure NORM vs. NORM with subclinical conduction findings. (b) Cohen's d effect sizes by feature. (c) ROC: pure NORM vs. subclinical (AUC $= 0.754$).

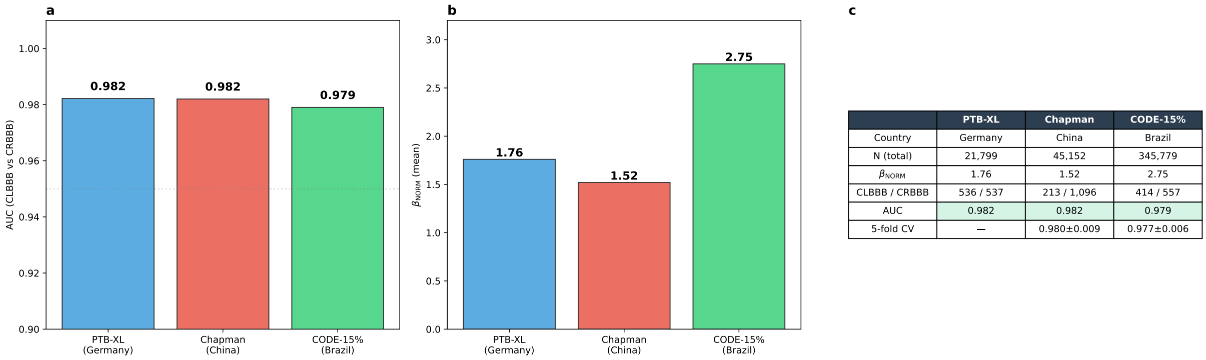
3.5 External Validation Across Three Continents

We applied the identical IRASA pipeline to two independent external datasets (Figure 5, Table 1).

Table 1: Cross-population replication of CLBBB vs. CRBBB classification.

Dataset	Country	Equipment	N	CLBBB	CRBBB	β_{NORM}	AUC
PTB-XL	Germany	Schiller	21,799	536	537	1.76	0.982
Chapman	China	GE MUSE	45,152	213	1,096	1.52	0.982
CODE-15%	Brazil	TEB	345,779	414	557	2.75	0.979

The absolute value of β_{NORM} varied substantially (1.52–2.75), reflecting equipment differences. However, the CLBBB vs. CRBBB AUC was virtually identical (0.982, 0.982, 0.979), demonstrating that the *relative geometry* of the twelve-lead β-vector is invariant to equipment and population. Spatial fingerprints replicated across all three datasets. The aging trajectory did not replicate externally ($\rho = +0.12$ in Chapman, $\rho = 0.005$ in CODE), likely reflecting absence of curated healthy cohorts and large equipment-related β-offsets.



	PTB-XL	Chapman	CODE-15%
Country	Germany	China	Brazil
N (total)	21,799	45,152	345,779
β_{NORM}	1.76	1.52	2.75
CLBBB / CRBBB	536 / 537	213 / 1,096	414 / 557
AUC	0.982	0.982	0.979
5-fold CV	—	0.980±0.009	0.977±0.006

Figure 5: **External validation across three continents.** (a) CLBBB vs. CRBBB AUC across datasets. (b) β_{NORM} varies by equipment but the classification geometry is invariant. (c) Cross-dataset summary.

3.6 Mortality Analysis

In CODE-15% ($n = 23{,}833$), higher β_{mean} was associated with mortality in unadjusted analysis (HR $= 1.79$, 95% CI 1.48–2.18, $p = 4.2 \times 10^{-9}$). After adjustment for age and sex, the association attenuated (HR $= 1.23$, $p = 0.019$). After full adjustment including diagnostic labels, the association was abolished (HR $= 1.02$, $p = 0.83$). β is a diagnostic marker, not an independent prognostic biomarker.

4 Discussion

4.1 The Spectral Exponent as a Window onto Cardiac Electrophysiology

We have shown that a single physics-based parameter—the spectral exponent β of the aperiodic ECG component—carries diagnostic information across the full clinical spectrum, from subclinical conduction abnormalities through healthy aging to overt bundle branch block. The key insight is not that β is a superior classifier, but that β distills a specific physical property: the balance between organized and disorganized broadband activity. This interpretability is the main advantage over black-box approaches.

4.2 Three Regimes of Cardiac Electrical Complexity

Our results organize into three regimes. In the *subclinical regime*, incomplete conduction disturbances fragment the wavefront without a dominant alternative pathway—more high-frequency fluctuation, lower β. In the *normal regime*, coordinated wavefronts yield $\beta \approx 1.76$. In the *overt blockade regime*, a major pathway is interrupted, forcing slow conduction through poorly coupled myocardium—steeper spectrum, higher β. Healthy aging follows the trajectory toward the subclinical regime, reflecting diffuse interstitial fibrosis and gap junction remodeling [de Jong et al., 2011, Jansen et al., 2010]. The breakpoint near age 53 coincides approximately with the onset of accelerated age-related ventricular remodeling [Cheng et al., 2009].

4.3 Cross-Population Invariance: Spectral Anatomy as a Physical Constant

The near-identical CLBBB vs. CRBBB AUC across three datasets (0.982, 0.982, 0.979)—spanning three continents, recording systems, and populations—is the central result. This invariance persists despite factor-of-two variation in absolute β-values, demonstrating that absolute β is equipment-dependent but the *relative geometry* of the β-vector is determined by anatomy. The physical explanation is straightforward: the left bundle passes through the interventricular septum beneath V1–V4 in every human heart, and the right bundle passes close to V1. These anatomical relationships are invariant. The twelve-lead β-vector is a low-dimensional spectral image of conduction system integrity.

4.4 Comparison with Prior Work

The HRV literature has extensively characterized $1/f$ scaling in RR-interval fluctuations [Peng et al., 1995, Ivanov et al., 1999, Goldberger et al., 2002]. Our work extends the fractal dynamics framework to the ECG waveform itself—a fundamentally different signal reflecting collective myocardial depolarization rather than autonomic modulation. The healthy $\beta \approx 1.76$ (near Brownian noise) contrasts with healthy HRV $\alpha \approx 1.0$ (pink noise), consistent with different physiological mechanisms. Schmidt et al. [2025] showed ECG spectral slope flattens with age using IRASA on $\sim$2,300 recordings. We confirm and extend this on a larger dataset and additionally demonstrate diagnostic specificity, the aging-pathology bifurcation, and cross-population invariance. Contoyiannis et al. [2021] found MI patients are "far from criticality"—conceptually aligned with our observation. George et al. [2025] achieved AUC = 0.90 on PTB-XL with Higuchi fractal dimension; our approach provides the spatial anatomical mapping as central contribution.

4.5 Limitations and Calibration of Claims

Replicated across three datasets: diagnostic β-landscape, spatial fingerprints, CLBBB vs. CRBBB classification, diagnostic ordering. These are our strongest claims. **PTB-XL only:** aging bifurcation, beat-to-beat variability, subclinical detection. The aging trajectory did not replicate externally ($\rho = +0.12$ in Chapman, $\rho = 0.005$ in CODE-15%—opposite or null). This is likely because (i) equipment-related β-offsets (β_{NORM} ranges from 1.52 to 2.75) dominate the subtle aging signal ($\Delta\beta \approx 0.1$ per decade), and (ii) external datasets lack curated healthy cohorts with confirmed absence of pathology. The aging findings should therefore be considered a PTB-XL observation requiring independent validation, not a confirmed cross-population result. **Honest null results:** β does not independently predict mortality (adjusted HR = 1.02). Absolute β is equipment-dependent. β does not correlate with age in uncurated populations. **Subclinical detection** is hypothesis-generating: AUC = 0.754 in a single dataset with a small sample

($n = 507$). External validation on datasets with annotated subclinical conduction findings is needed before clinical conclusions can be drawn. **Relationship to standard morphological features:** β correlates with QRS duration ($\rho = 0.24$, $p < 10^{-68}$), as expected since bundle branch block both widens the QRS and steepens the spectrum. However, β carries substantial information beyond QRS duration: for CLBBB vs. CRBBB classification, QRS duration alone achieves AUC = 0.688 while the β-vector achieves AUC = 0.988 (ΔAUC = +0.300). The diagnostic effect of β on CD vs. NORM retains 96% of its magnitude after residualizing on QRS duration (Cohen's $d = 0.90$ vs. 0.93 raw). This indicates that β captures spatial conduction properties—the twelve-lead geometry of spectral slopes—not merely QRS width. Additional limitations: (1) ten-second duration; (2) cross-sectional design; (3) no medication data; (4) inability to separate autonomic from intrinsic myocardial effects; (5) wearable utility (reduced leads) untested.

4.6 Clinical Implications and Future Directions

The cross-equipment invariance suggests β-based characterization could be deployed across heterogeneous environments without device-specific recalibration. However, β does not replace existing methods. Its value lies in: (1) interpretability; (2) equipment-invariance of relative patterns; and (3) computability without waveform delineation. Whether these translate to clinical utility in wearable screening, longitudinal monitoring, or telemedicine remains to be determined. Priorities include single-lead validation, longitudinal β-tracking, and investigation of the molecular substrate using single-cell transcriptomic data from aging hearts [Tabula Muris Consortium, 2020].

Ethics Statement

This study exclusively uses publicly available, de-identified datasets previously approved for research use by institutional review boards at the originating institutions: PTB-XL (Physikalisch-Technische Bundesanstalt, Berlin), Chapman-Shaoxing (Shaoxing People's Hospital), and CODE-15% (Telehealth Network of Minas Gerais, approved by the Research Ethics Committee of the Universidade Federal de Minas Gerais). No additional ethical approval was required for this secondary analysis of de-identified data.

Author Contributions

T.S. conceived the study, designed and performed all analyses, generated all figures, and wrote the manuscript.

Data and Code Availability

All data are publicly available: PTB-XL (`https://physionet.org/content/ptb-xl/`), Chapman-Shaoxing (`https://physionet.org/content/ecg-arrhythmia/`), CODE-15% (`https://zenodo.org/records/4916206`). Analysis code is available as a Kaggle notebook: `https://www.kaggle.`

com/code/theodorspiro/the-heartbeat-at-the-edge-of-chaos and on GitHub: https://github.com/mool32/ecg-spectral-exponents.

Declaration of Interests

The author has no financial or non-financial competing interests to declare.

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References

Babayan A, et al. A mind-brain-body dataset of MRI, EEG, cognition, emotion, and peripheral physiology in young and old adults. *Scientific Data*. 2019;6:180308.

Cheng S, Xanthakis V, Sullivan LM, et al. Correlates of echocardiographic indices of cardiac remodeling over the adult life course. *Circulation*. 2009;119:2563–2571.

Contoyiannis Y, Stavrinides SG, Hanias MP, et al. Can high-frequency ECG fluctuations differentiate between healthy and myocardial infarction cases? *Results in Physics*. 2021;25:104169.

de Jong S, van Veen TAB, van Rijen HVM, de Bakker JMT. Fibrosis and cardiac arrhythmias. *J Cardiovasc Pharmacol*. 2011;57(6):630–638.

Gao R, Peterson EJ, Voytek B. Inferring synaptic excitation/inhibition balance from field potentials. *NeuroImage*. 2017;158:70–78.

George T, et al. Exploring complexity changes in diseased ECG signals for enhanced classification. *arXiv:2510.17810*. 2025.

Goldberger AL, Amaral LAN, Hausdorff JM, et al. Fractal dynamics in physiology: alterations with disease and aging. *Proc Natl Acad Sci USA*. 2002;99(suppl 1):2466–2472.

Ivanov PCh, Amaral LAN, Goldberger AL, et al. Multifractality in human heartbeat dynamics. *Nature*. 1999;399:461–465.

Jansen JA, van Veen TAB, de Bakker JMT, van Rijen HVM. Cardiac connexins and impulse propagation. *J Mol Cell Cardiol*. 2010;48(1):76–82.

Lendner JD, Helfrich RF, Mander BA, et al. An electrophysiological marker of arousal level in humans. *eLife*. 2020;9:e55092.

Lipsitz LA, Goldberger AL. Loss of 'complexity' and aging. *JAMA*. 1992;267(13):1806–1809.

Peng C-K, Havlin S, Stanley HE, Goldberger AL. Quantification of scaling exponents and crossover phenomena in nonstationary heartbeat time series. *Chaos*. 1995;5(1):82–87.

Ribeiro AH, Ribeiro MH, Paixão GMM, et al. Automatic diagnosis of the 12-lead ECG using a deep neural network. *Nat Commun*. 2020;11:1760.

Schmidt H, Bielczyk NZ, Krasnozhon A, et al. Age-related changes in "cortical" 1/f dynamics are linked to cardiac activity. *eLife*. 2025;13:RP102750.

Shekatkar SM, Kotriwar Y, Harikrishnan KP, Ambika G. Detecting abnormality in heart dynamics from multifractal analysis of ECG signals. *Sci Rep*. 2017;7:15127.

Strodthoff N, Wagner P, Schaeffter T, Samek W. Deep learning for ECG analysis: benchmarks and insights from PTB-XL. *IEEE J Biomed Health Inform*. 2021;25(5):1519–1528.

Tabula Muris Consortium. A single-cell transcriptomic atlas characterizes ageing tissues in the mouse. *Nature*. 2020;583:590–595.

Voytek B, Kramer MA, Case J, et al. Age-related changes in 1/f neural electrophysiological noise. *J Neurosci*. 2015;35(38):13257–13265.

Wagner P, Strodthoff N, Bousseljot R-D, et al. PTB-XL, a large publicly available electrocardiography dataset. *Sci Data*. 2020;7:154.

Wen H, Liu Z. Separating fractal and oscillatory components in the power spectrum. *Brain Topogr*. 2016;29(1):13–26.

Yang J, Ning X, He A. Identification of healthy and pathological heartbeat dynamics based on ECG-waveform using multifractal spectrum. *Physica A*. 2020;559:125021.

Zheng J, Zhang J, Danioko S, et al. A 12-lead electrocardiogram database for arrhythmia research covering more than 10,000 patients. *Sci Data*. 2020;7:48.

Supplementary Information

Table S1. Pairwise superclass comparisons

Comparison	n_1	n_2	$\bar{\beta}_1$	$\bar{\beta}_2$	Cohen's d	p (Bonferroni)
NORM vs CD	9,068	1,708	1.760	2.151	0.989	$< 10^{-120}$
STTC vs CD	2,400	1,708	1.768	2.151	0.760	$< 10^{-80}$
MI vs CD	2,532	1,708	1.859	2.151	0.577	$< 10^{-40}$
HYP vs CD	534	1,708	1.859	2.151	0.501	$< 10^{-16}$
NORM vs HYP	9,068	534	1.760	1.859	0.293	3.0×10^{-7}
NORM vs MI	9,068	2,532	1.760	1.859	0.281	$< 10^{-23}$
STTC vs HYP	2,400	534	1.768	1.859	0.231	6.6×10^{-6}
MI vs STTC	2,532	2,400	1.859	1.768	−0.227	$< 10^{-13}$
NORM vs STTC	9,068	2,400	1.760	1.768	0.023	1.0 (n.s.)
MI vs HYP	2,532	534	1.859	1.859	0.002	1.0 (n.s.)

Single-label assignments used. p-values Bonferroni-corrected for 10 comparisons.

Table S2. Per-subtype CD metrics

Subtype	n	$\beta_{\text{mean}} \pm$ SD	Cohen's d vs NORM
CLBBB	536	2.923 ± 0.394	3.438
CRBBB	537	2.142 ± 0.490	1.107
IVCD	780	2.131 ± 0.482	1.064
LAFB	1,604	2.070 ± 0.508	0.848
1AVB	779	2.062 ± 0.554	0.847
IRBBB	1,102	1.771 ± 0.377	0.033

Sorted by descending Cohen's d. Counts use $\geq$ 80% annotator confidence threshold.

Table S3. Segmented regression parameters

Metric	Breakpoint	Slope (left)	Slope (right)	F	p
β_{mean}	32 y	+0.0046/y	−0.0039/y	13.5	1.4×10^{-6}
σ_β	53 y	+0.00036/y	+0.00012/y	17.8	1.9×10^{-8}

Fitted on $n = 8{,}940$ NORM recordings (age 18–95). σ_β shows acceleration of spatial heterogeneity after age 53.

347 **Table S4. Demographics by diagnostic superclass**

Superclass	n	Male (%)	Age (mean $\pm$ SD)
NORM	9,514	46.1	52.1 ± 17.2
MI	5,469	62.3	66.3 ± 12.5
STTC	5,235	49.0	66.3 ± 13.5
CD	4,898	61.2	65.3 ± 15.0
HYP	2,649	57.4	65.9 ± 13.9
Total	21,799	52.1	59.5 ± 16.8

Multi-label counts (recordings may appear in multiple superclasses). One age outlier (300 y) excluded from age statistics.

348 **Figure S1. IRASA fit quality (R^2 distribution)**

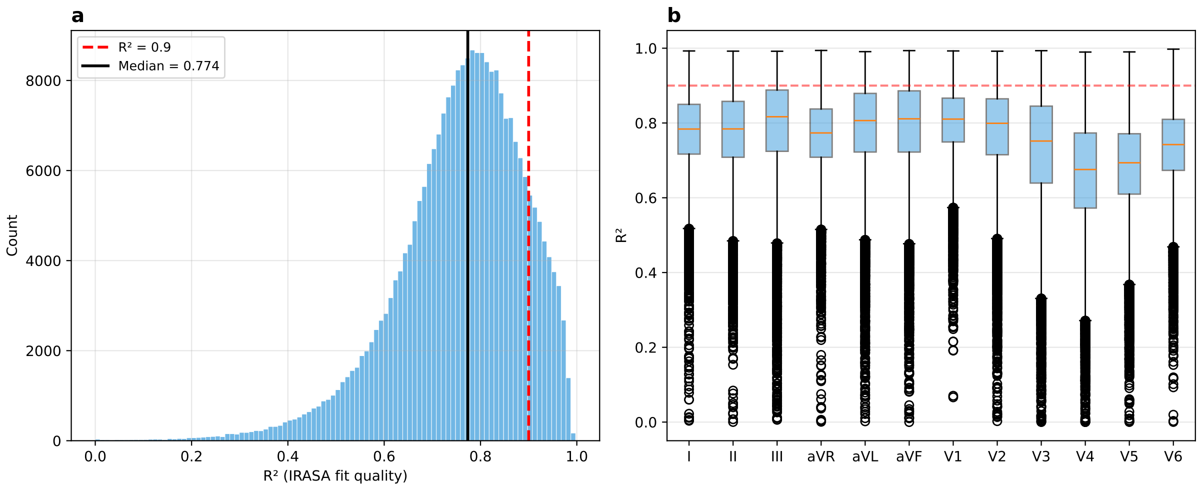

Figure 6: (a) Histogram of R^2 values across all leads and recordings (median = 0.774). (b) R^2 by lead: precordial leads V1–V2 show highest R^2, reflecting stronger $1/f$ structure near the septum.

Figure S2. Sensitivity analyses

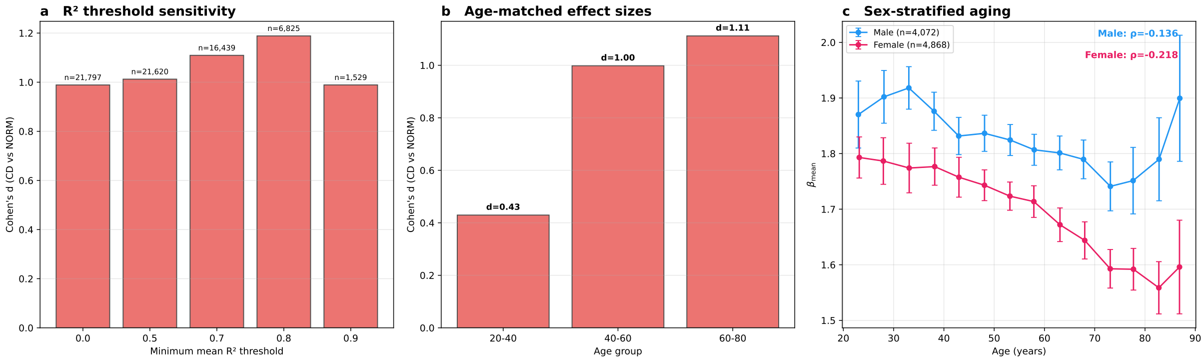

Figure 7: (a) CD vs. NORM effect size is robust to R^2 quality thresholds. (b) Age-matched subsamples confirm the diagnostic effect is not driven by age confounding. (c) Sex-stratified aging trajectories: women show steeper decline ($\rho = -0.218$ vs. -0.136 for men).